

Thermal anomaly EVT-IN-ODI-8364F540 is classified as IND\_FLARE with 91.5% calibrated model confidence and severity tier ABNORMAL (Z = +2.8σ). Peak radiative output reached 11.97 MW across 16 verified multi-sensor satellite passes. Centroid mapped over Agricultural Cropland without direct plant boundary overlap.

12.0 MW  
PEAK RADIANCE

16 Passes  
SATELLITE DETECTIONS

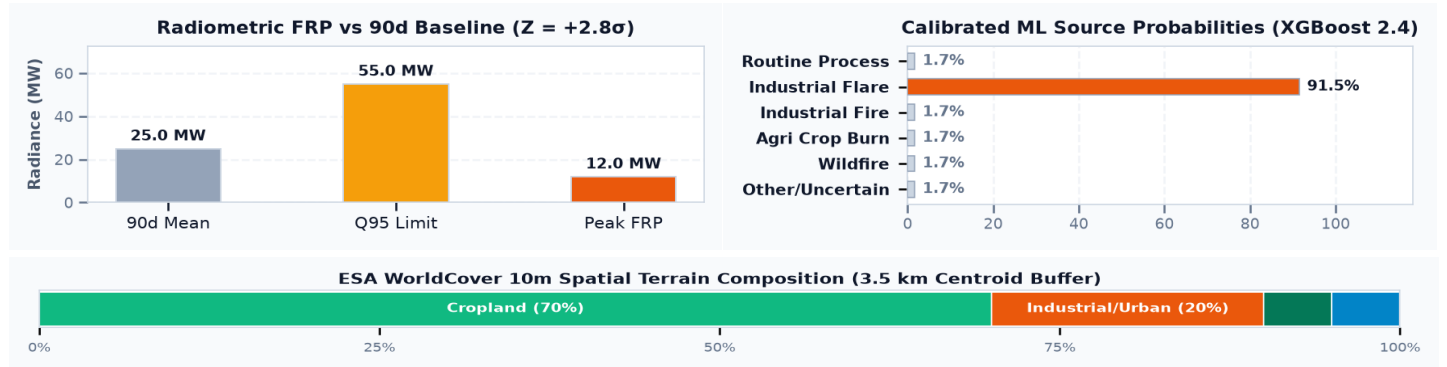
91.5%  
CALIBRATED SOFTMAX

+2.8σ  
ANOMALY DEVIATION

1. Verified Multi-Sensor Radiometric Metrics

| Metric Name                | Observed Telemetry            | Historical 90d Baseline             | Radiometric Assessment               |
|----------------------------|-------------------------------|-------------------------------------|--------------------------------------|
| Peak Radiative Power (FRP) | 11.97 MW                      | 25.00 MW (Facility Mean)            | 0.5x Baseline Mean                   |
| Mean Cluster Radiance      | 3.16 MW                       | Persistence: TRANSIENT              | Z-Score: +2.8σ Statistical Departure |
| Max Brightness Temp (BT)   | 352.4 K                       | Sensor Bands: VIIRS I4 (375m) / M13 | High-Intensity Combustion Confirmed  |
| Temporal Detection Window  | 2026-08-26T19:13 to 08:07 UTC | 16 Sensor Passes Ingested           | Active Multi-Sensor Track            |

2. Visual Radiometric Analytics & ML Attribution (High-DPI)



3. Centroid Location & Associated Industrial Plant Audit

| Attribute            | Forensic Detail                   | Attribute               | Forensic Detail                  |
|----------------------|-----------------------------------|-------------------------|----------------------------------|
| Centroid Coordinates | 20.79344°N, 85.25465°E            | Primary Land Use        | Agricultural Cropland            |
| State / Territory    | Odisha                            | District / Jurisdiction | Angul                            |
| Associated Facility  | No Direct Single-Facility Overlap | Spatial Buffer          | Open Industrial / Rural Corridor |

4. Nearby Sovereign Industrial Infrastructure (25 km Search Radius)

| Facility Name | Sector Category  | State / District | Radial Distance | Baseline FRP |
|---------------|------------------|------------------|-----------------|--------------|
| KAMALANGA     | Power Generation | India            | 8.6 km          | 0.0 MW       |
| TALCHER       | Power Generation | India            | 13.8 km         | 0.0 MW       |

5. Regional Active Anomaly Cluster & Concurrent Events (75 km Buffer)

| Event Ref           | Classification | Severity Tier | Peak Radiance | Distance | Latest Detection (UTC) |
|---------------------|----------------|---------------|---------------|----------|------------------------|
| EVT-IN-ODI-15BE1E73 | IND_ROUTINE    | NORMAL        | 2.6 MW        | 1.9 km   | 2026-08-26 20:53       |
| EVT-IN-ODI-44BF81A4 | IND_ROUTINE    | NORMAL        | 2.1 MW        | 5.1 km   | 2026-08-26 19:15       |
| EVT-2026-15EF29     | IND_ROUTINE    | NORMAL        | 9.7 MW        | 20.2 km  | 2026-08-27 08:07       |
| EVT-IN-ODI-7DBE808F | IND_ROUTINE    | NORMAL        | 9.0 MW        | 20.7 km  | 2026-08-27 07:09       |

6. Multi-Sensor Satellite Telemetry & Radiometric Pass Register

| # | Detection Time (UTC) | Sensor Platform | Radiant Output | Brightness (K) | Day/Night  |
|---|----------------------|-----------------|----------------|----------------|------------|
| 1 | 2026-08-26T19:13     | N20             | 1.95 MW        | 303.9 K        | Night Pass |
| 2 | 2026-08-26T19:15     | N20             | 2.09 MW        | 301.9 K        | Night Pass |
| 3 | 2026-08-26T19:57     | N21             | 0.78 MW        | 306.6 K        | Night Pass |
| 4 | 2026-08-26T20:34     | N               | 1.13 MW        | 302.2 K        | Night Pass |
| 5 | 2026-08-26T20:34     | N               | 1.13 MW        | 303.8 K        | Night Pass |

7. Statutory SOP Compliance Directives & Containment Protocol

|                                                                                                                                                                                             |                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>ACTION 01: Immediate On-Site Physical Inspection:</b> Dispatch SPCB / Regional Disaster Response team to verify combustion source and evaluate containment perimeter.                    |                                 |
| <b>ACTION 02: Continuous Emission Telemetry (CEMS) Audit:</b> Cross-verify Continuous Emission Monitoring Systems data and industrial flaring logs against satellite radiance timestamps.   |                                 |
| <b>ACTION 03: Safety Buffer &amp; Containment Directive:</b> Enforce active 2.5 km industrial safety perimeter and initiate cooling operations if thermal anomaly persistence exceeds 4.0σ. |                                 |
| CLEARANCE: OFFICIAL NATIONAL SURVEILLANCE DOSSIER // RESTRICTED ACCESS                                                                                                                      | INTEGRITY: SHA256-AUTHENTICATED |